

Chapter 15

Game Theory

Learning Objectives

- 15.1 Understand the use of game theory to analyze situations where the profit of one firm depends critically on the actions of others.
- 15.2 Distinguish between sequential-move games and simultaneous-move games.
- 15.3 Analyze the likely outcome of a game known as Nash equilibrium.
- 15.4 Identify equilibria in different types of games and how to change the rules of the game to your advantage.

Chapter 15 – Main Points

- A **Nash equilibrium** is a pair of strategies, one for each player, in which each strategy is a best response against the other.
- When players act rationally, optimally, and in their own self-interest, it's possible to compute the likely outcomes of games. By studying games, we learn where the pitfalls are and how to avoid them.
- Sequential games include a potential first-mover advantage, or disadvantage, and players can change the outcome by committing to a future course of action.. Credible commitments are difficult to make because they require that players threaten to act in an unprofitable way—against their self-interest.
- In **simultaneous-move games**, players move at the same time.
- In the **prisoners' dilemma**, conflict and cooperation are in tension—self-interest leads to outcomes that no one likes. Studying the games can help you figure a way to avoid these bad outcomes.
- In repeated games, it is much easier to get out of bad situations. Here are some general rules of thumb:
 - Be nice: No first strikes.
 - Be easily provoked: Respond immediately to rivals.
 - Be forgiving: Don't try to punish competitors too much.
 - Don't be envious: Focus on your own slice of the profit pie, not on your competitor's.
 - Be clear: Make sure your competitors can easily interpret your actions.

Warm-Up

- Think of a time when what you chose depended on what someone else was going to do, but you had to decide at the same time. What happened?

Game Theory Basics

- Types of Games

- Zero-sum game: offsetting gains/losses.
- Positive sum game: potential for mutual gain.
- Negative-sum game: potential for mutual loss.
- Cooperative games: joint action is favored.
- Repeated games: games that occur over multiple periods

- Role of Interdependence

- Sequential games involve successive moves.
- Simultaneous-move games incorporate coincident moves.

Simultaneous Game

- Simultaneous games are easily represented by using a payoff matrix
 - The game is laid out in table format
- The simplest games involve:
 - Two players
 - Two strategies
 - A strategy is a move the player can take
 - And are one-shot games (they only occur once)
 - Players know their payoffs and may or may not know the other player's payoffs

Reading a Payoff Matrix

Player 1 has strategies of Up and Down
Player 2 has strategies of Left and Right

If Player 1 selects Up and Player 2 selects Left, then Player 1 gets a and Player 2 gets b

If Player 1 selects Down and Player 2 selects Right, then Player 1 gets g and Player 2 gets h

		Player 2	
		Left	Right
Player 1	Up	a , b	c , d
	Down	e , f	g , h

Profits for Firms

- In the following game, each firm considers the strategies of pricing high or low and the payoffs are the profit for each strategy.

		Firm 2	
		Price High	Price Low
Firm 1	Price High	6 , 1	4 , 3.5
	Price Low	2 , 2	3 , 3

- **Dominant strategy** – each player chooses an action that allows them the best outcome regardless of the other player's actions.
 - Firm 1 has a dominant strategy to always price high since $6 > 2$ and $4 > 3$
 - Firm 2 has a dominant strategy to always price low since $3.5 > 1$ and $3 > 2$

Prisoner's Dilemma

- Classic Riddle
 - Rational individual behavior can give suboptimal group result.
 - Rationality can hamper beneficial cooperation.
- Business Application
 - Dominant strategy gives best result regardless of moves by other players.
 - Secure strategy gives best result assuming the worst possible scenario.

Pepsi and Coke

		Pepsi	
		Lower Price	Keep Price
Coke	Lower Price	42 , 39	45 , 36
	Keep Price	39 , 42	43 , 40

Coke has a dominant strategy to lower price, and Pepsi has a dominant strategy to lower price

The result is suboptimal since the combined payoff (81) is less than if they both chose to keep price (83)

Prisoner's Dilemma

Consider 2 people who were caught doing something illegal. If one confesses and the other stays quiet, then the silent one goes to jail for 9 years, with the opposite being true. If they both confess, then each gets 6 years. If they both remain silent then each gets 1 year on a technicality.

		Prisoner 2	
		Confess	Stay Quiet
Prisoner 1	Confess	-6 , -6	0 , -9
	Stay Quiet	-9 , 0	-1 , -1

Rational actors will both choose to Confess

Discussion Activity 1

- If you were Pepsi or Coke and knew you were in a Prisoner's Dilemma situation, what would you do?
- Consider price matching, advertising coordination, or long-term contracts. How might repeated interaction change your thinking?

Nash Equilibrium

- Nash Equilibrium Concept
 - Neither player can improve their payoff through a unilateral change in strategy.
 - In the Prisoner's dilemma, choosing to Confess by both players was a Nash equilibrium.
 - Nash equilibrium concept is broader than the concept of a dominant strategy equilibrium.
 - Every dominant strategy equilibrium is also a Nash equilibrium.
 - Nash equilibrium can exist where there is no dominant strategy equilibrium.
- Suppose Delta and American both face pressure to cut prices on a competitive route.
 - The Prisoner's Dilemma dynamic explains why they often end up in price wars despite mutual harm.

More Games

		Pepsi	
		Lower Price	Keep Price
Coke	Lower Price	1 , 1	0 , 0
	Keep Price	0 , 0	2 , 2

Two Nash equilibria emerge: Lower Price, Lower Price and Keep Price, Keep Price

Which do you choose, or is there an alternative?

Credibility

- It may often be the case that one firm has a dominant strategy and the other does not. If the latter firm wants to get a particular payoff it would need to have some type of credible threat (e.g. an announcement)

		Pepsi	
		Lower Price	Keep Price
Coke	Lower Price	10 , 8	15 , 6
	Keep Price	4 , 6	6 , 10

- To get the outcome Pepsi wants they could say they would lower their prices by 10% more than Coke.

The Free Riding Dilemma

- An MBA study group is assigned a project. Each member can work hard or shirk. If both work hard, they get A's; if both shirk, they get C's; if only one works hard, they both get B's.
- Suppose their preferences are identical:
 1. Shirk and a B is preferred to
 2. Work and an A which is preferred to
 3. Shirk and a C, which is preferred to
 4. Work and a B.

The Free Riding Dilemma

		Student 2	
		Shirk	Work
Student 1	Shirk	C + Leisure, C + Leisure	B + Leisure, B
	Work	B, B + Leisure	A, A

- The Nash equilibrium is (shirk, shirk) – BUT, as any student will verify, this outcome is inefficient, the students would jointly *prefer* {work, work} outcome (and the resulting “A”).
- This outcome is not an equilibrium though, once your partner is working hard, the best response is to shirk.
- A successful study group finds a way to manage this conflict and avoid the {shirk, shirk} equilibrium

Indifference

What if a firm is indifferent between two payouts?

		Pepsi	
		Lower Price	Keep Price
Coke	Lower Price	10 , 7	15 , 7
	Keep Price	4 , 8	6 , 6

Coke has a dominant strategy to Lower Price, but Pepsi is now indifferent at that strategy when selecting to Lower Price or Keep Price.

If Pepsi and Coke are rivals, then Pepsi may lower price to keep Coke from earning more profit.

Beyond 2 Strategies

- In many cases firms are faced with more than two strategies
- For example, a firm may choose to raise, lower, or keep its price/output the same.
- Likewise, a competing firm has the same strategies.
- Solving these games requires **eliminating dominated strategies**

3 Strategy Game

		Pepsi		
		Lower Price	Keep Price	Raise Price
Coke	Lower Price	4 , 3	5 , 1	6 , 2
	Keep Price	2 , 1	8 , 4	3 , 6
	Raise Price	3 , 0	9 , 6	2 , 8

Eliminate dominated strategies.

1. Pepsi can eliminate Keep Price as Lower Price or Raise Price always dominate it
 1. Reform the matrix removing Keep Price for Pepsi
2. Coke can eliminate Keep Price and Raise Price as Lower Price dominates them
3. Coke chooses Lower Price and Pepsi chooses Lower Price

Activity – Design Your Own Game

Scenario:

- You're advising two rival delivery platforms in a competitive metro market. They can choose to either cut prices, maintain prices, or offer loyalty rewards in the next quarter.

Instructions:

1. Create a 3×3 payoff matrix showing the strategies and expected profits for each firm.
2. Identify if either firm has a dominant strategy.
3. Determine any Nash equilibria.
4. If no clear equilibrium exists, propose a way to alter the game (e.g., long-term contracts, signaling) to reach a better outcome.

Activity – Design Your Own Game

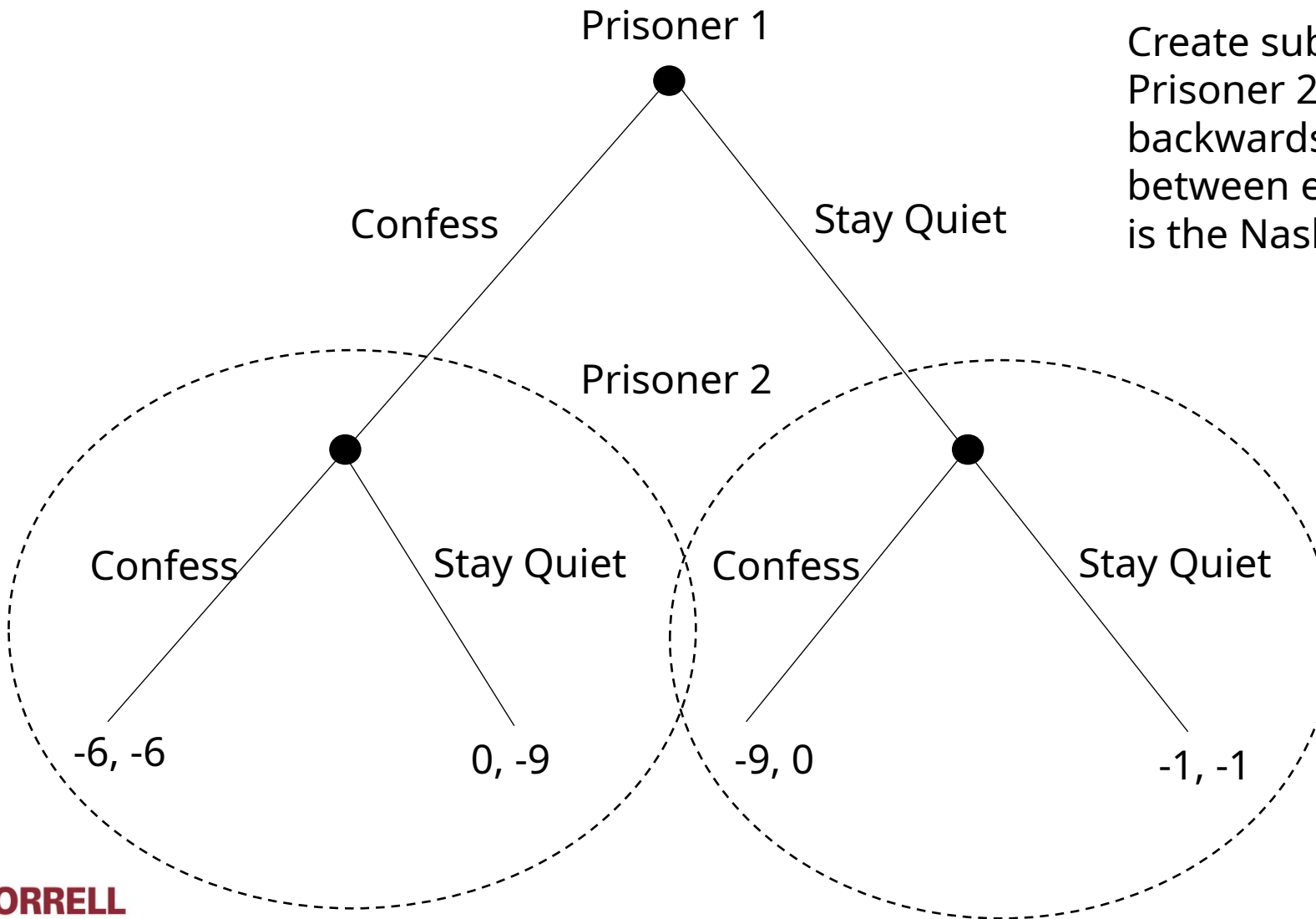
		Pepsi		
		Lower Price	Keep Price	Raise Price
Coke	Lower Price	2 , 2	4 , 1	3 , 2.5
	Keep Price	1 , 4	3 , 3	2.5 , 3.5
	Raise Price	2.5, 3	3.5, 2.5	3 , 3

- [
- New game gives weakly dominant strategies of Raise Price, Raise Price
- Nash Equilibrium likely fall to (Raise Price, Raise Price)
- Game Alteration:
 - Firms could sign public agreements to hold pricing (signaling).
 - Implement tit-for-tat strategy in repeated quarters.
 - Build switching costs (e.g., reward programs) to stabilize payoffs.

Sequential Games

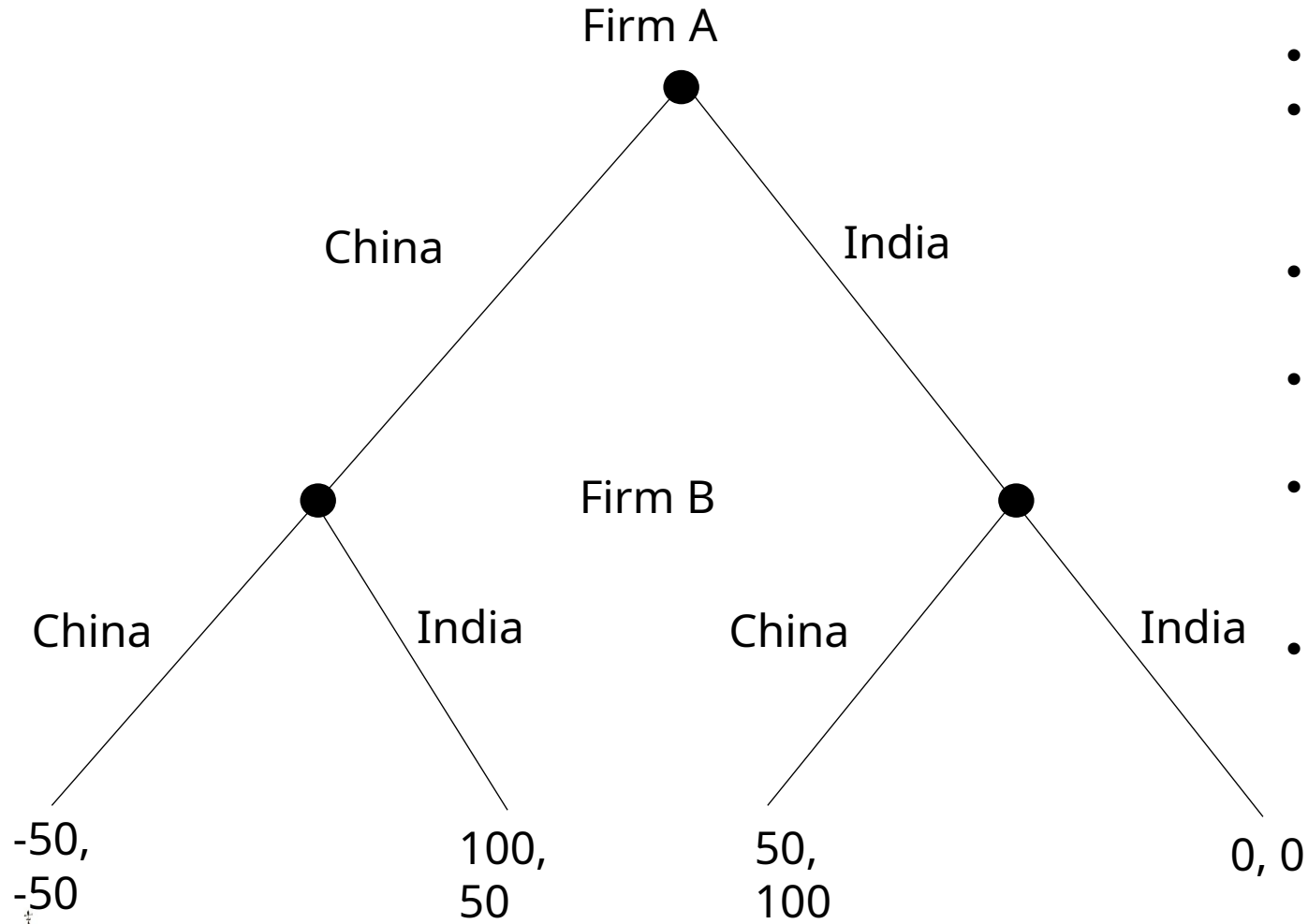
- Sequential games make use of the game tree.
- Each games starts with one player making a choice, and the second player may be able to or not able to observe the choice.
- To solve sequential games, we need to do **backwards induction**.
 - Backwards induction means we start at the end of the game and works towards the start

Prisoner's Dilemma



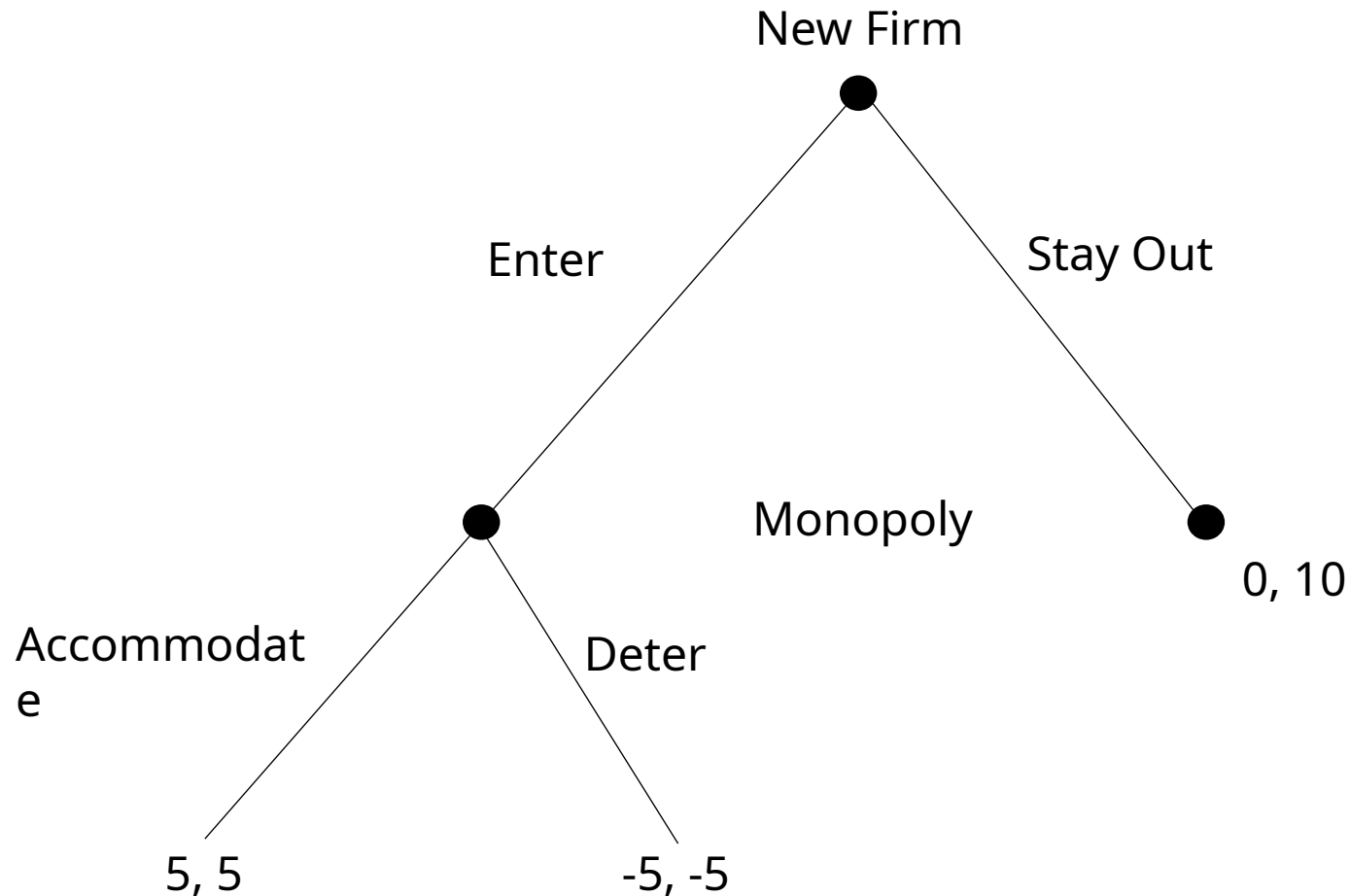
Create subgames at the game nodes. Prisoner 2 then solves the game via backwards induction. Compare the payoffs between each to see that Confess, Confess is the Nash Equilibrium.

Exploring Foreign Markets



- Create subgames at the separate nodes
- Firm A now uses backwards induction by comparing the payoffs to Firm B at each subgame
- On the left subgame, Firm B prefers to go to India since $50 > -50$
- On the right subgame, Firm B prefers to go to China since $100 > 0$
- Firm A compares the payoffs they receive from each subgame (100 on the left and 50 on the right)
- Firm A chooses China and Firm B chooses India

Entry Deterrence



- Create two subgames at the second set of nodes
- Compare monopoly profits on the left subgame
- Monopoly will choose accommodate
- New firm compares the 5 from the left subgame and 0 from the right subgame
- New firm should enter
- Could the monopoly successfully deter?

Game of Chicken

- This is a classic game in which two people are driving towards each other.
- If they collide, then both die
- If they both swerve to miss, then they both lose face
- If one swerves and the other stays true, then one loses face and the other looks strong, and vice versa
- There are two Nash equilibria in this game and that is for one to swerve and the other not alter their direction.
- The key to coming out the winner in this game is to make a hardline commitment to alter the strategies of the game (or obtain a first mover advantage)
- We can also place it as a battle of the sexes, where a husband and wife do not communicate beforehand about if they are going to meet at the movies or a restaurant. The husband goes to one and the wife to the other.

Game of Chicken

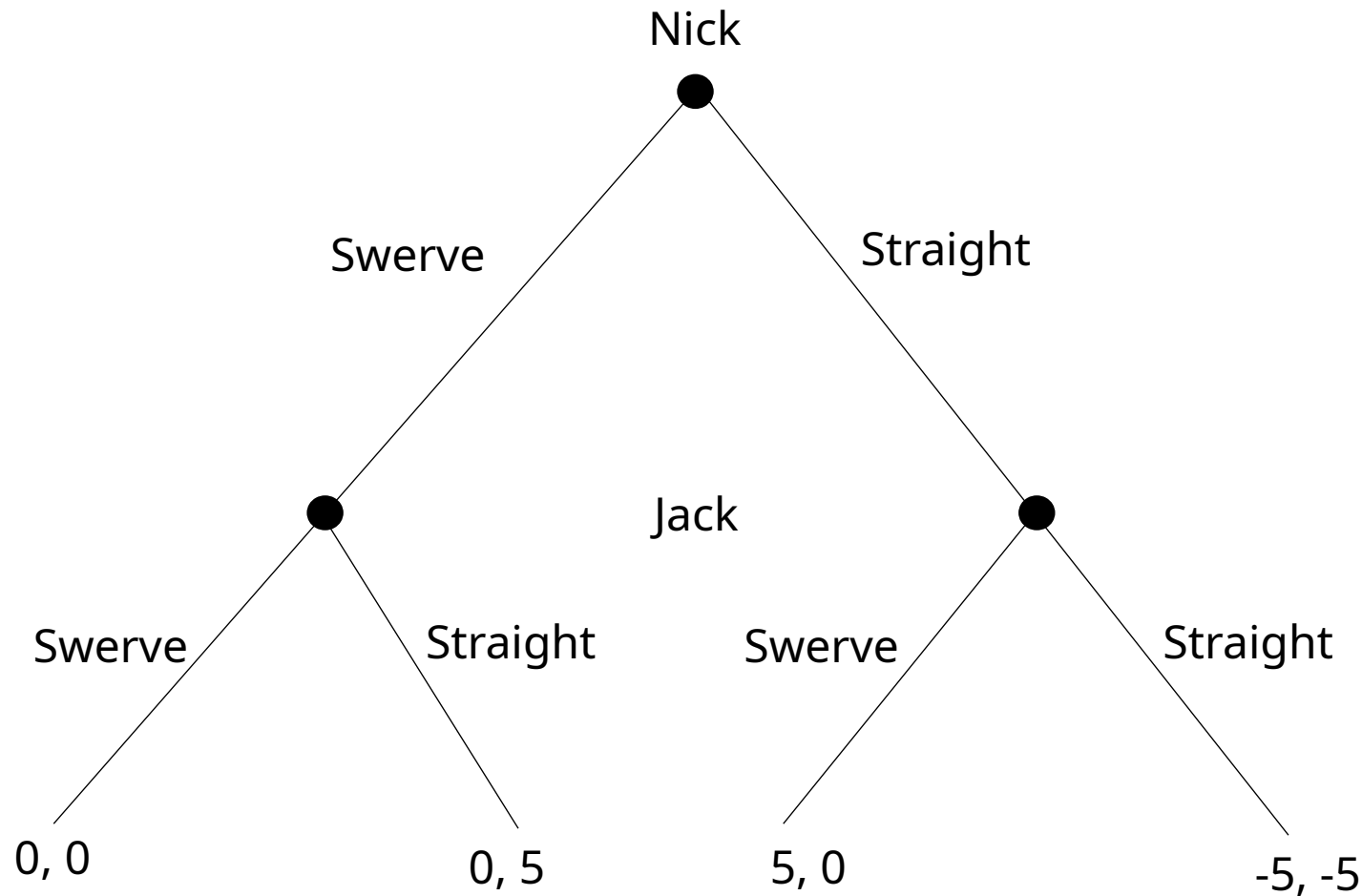
		Jack	
		Swerve	Stay Straight
Nick	Swerve	0 , 0	0 , 5
	Stay Straight	5 , 0	-5 , -5

In a simultaneous move game, Swerve, Swerve and Stay Straight, Stay Straight are Nash equilibria

How do we solve this?

- Commitment – one announces early they will commit to staying straight
- Mixed Strategy – set up probabilities for swerving or staying straight
- Sequential game – move to a sequential game

Game of Chicken



- Create two subgames at the second set of nodes
- Compare payoffs on the left and right subgames
 - $5 > 0$ and $0 > -5$
- Nick sees that Jack will always do the opposite of what he commits to
- Nick should choose to stay straight
- How can Nick make his commitment credible?
 - Reputation
 - Remove flexibility (yank the steering wheel)
 - Commit early and in full view
 - Delegate your decision to someone else

Game of Chicken

Consider Netflix and Disney+ both preparing to drop blockbuster releases on the same weekend.

- Only one can dominate headlines.
- Create a payoff matrix that captures these strategies and payoffs
 - If both Go Big, attention and profits may dwindle (1,1)
 - If one Holds Back, and one Go(es) Big the other wins the spotlight (8,1)
 - If both Hold(s) Back, then profits fall (0, 0)

What we find is that streaming platforms often play chicken with release dates or pricing knowing only one can dominate consumer attention.

Discussion Activity 2

- Think about Netflix and Disney+ planning big releases. How could they signal commitment not to back down?
- Think about early trailer drops, exclusive rights, or release dates. What makes a signal credible?

Repeated Games

- Repeated games take place over 2 to many rounds.
- But because we know there is a last round, we can solve the game through backwards induction.
- Suppose a game is played 10 times.
- The socially desirable outcome is for both to keep price (total payout of 10).
- But since the game ends at round 10, we know they will both revert to keeping price as one will deviate to get the higher payout and the other retaliates. This is a **trigger strategy** – strategies that cause you to react to certain actions taken by the other player
- Through backwards induction, we know that since the game ends in 10 rounds both revert to choosing to lower price

		Pepsi	
		Keep Price	Lower Price
Coke	Keep Price	5 , 5	1 , 8
	Lower Price	8 , 1	2 , 2

Repeated Games

- For airlines, collusive-like pricing stability (e.g. matching fares on key routes) can be sustained as long as no airline defects, backed by a threat of retaliation.
- For ride-share platforms, drivers or platforms may hold pricing truce until someone lowers fares, triggering immediate matched cuts.
- Firms in ride-sharing and airlines often rely on tacit coordination sustained by trigger strategies, where one defection resets the whole pricing dynamic.
 - But these firms must be careful as tacit collusion is illegal as [several airlines found out in 2015](#)

Infinitely Repeated Games

- Role of Reputation
 - Infinitely repeated games occur over and over again without boundary or limit.
 - Firms receive sequential payoffs that shape current and future strategies.
 - Reputations for high quality give consumers confidence for repeat transactions.
- How are these games solved?
 - The Nash Equilibrium concept still applies, but it can occur numerous times
 - Unlike repeated games where we have a last round, typically, the Pareto optimal outcome is achieved (the one with the highest payout)
 - Under this concept, grim strategies – where a player defects once and then is punished forever – doesn't hold.
 - Instead, tit-for-tat strategies are likely to arise.

Tit-for-Tat Strategy

- To solve infinitely repeated games, we employ a tit-for-tat strategy proposed by Robert Axelrod in 1980.
- The tit-for-tat strategy is one where you retaliate if someone deviates from the cooperative outcome (though cooperation amongst firms is illegal)
 - You begin by cooperating with your opponent.
 - If you or your opponent deviate from what is established, then your opponent or you retaliate the next round and cause more harm. This punishes the deviating opponent and brings back the cooperative actions.
- As your book states:
 - Be nice: no first strikes
 - Be forgiving don't punish competitors too much if they defect from the cooperative outcome
 - Be easily provoked: respond *immediately* to rivals.
 - Don't be envious: focus on your strategy and market share
 - Be clear: make sure your competitor can easily interpret your actions.

An Extension of the Infinitely Repeated Game

- Product Quality Game
 - Sellers choose whether to offer low or high quality
 - Buyers observe quality only after they purchase
 - If the game ends soon, then sellers will offer low quality
 - If the game is infinitely repeated, then sellers have the incentive to maintain quality to maintain profits

Discussion Activity 3

- Airlines often match prices quickly when a competitor cuts fares. Why don't they just undercut permanently?
- Consider “trigger strategies” and the threat of retaliation in repeated games.

Case Study: Airline Price Wars and Trigger Strategies

- Airlines operate in high fixed-cost, low marginal-cost industries.
- Most U.S. domestic routes are served by a few major players (e.g., Delta, American, United, Southwest).
- Pricing is highly interdependent; one firm's fare change directly affects rivals' profits.

Scenario: American Airlines Fare War (Early 2000s)

- American Airlines dropped prices aggressively on key routes (Dallas–Chicago, LA–NYC) to gain market share.
- Competitors quickly matched prices, triggering a fare war.
- Result: All carriers saw profits plunge, and American's market share gains were short-lived.
- Eventually, prices stabilized, suggesting tacit collusion had returned through trigger strategies.

Case Study: Airline Price Wars and Trigger Strategies

Why Airlines Don't Undercut Permanently

- Short-term gain often followed by long-term loss due to retaliation.
- Customers become trained to wait for deals, hurting long-run revenue.
- Without coordination, Prisoner's Dilemma results:
 - Undercut → rivals match → everyone loses.
- Trigger strategies create discipline without explicit collusion (which is illegal).

Lessons and Implications

- Game theory helps explain why cooperative behavior emerges without contracts.
- Airlines use tools like public fare announcements and capacity planning to signal intentions.
- Sustaining cooperation depends on:
 - Frequency of interaction (repeated games)
 - Monitoring ability (public pricing)
 - Credibility of punishment

Learning from Games

- If you seek to avoid the Prisoner's Dilemma, then alter the payoff structure of the game
- Ways to do this (changing your long-run strategy):
 - Make yourself less dependent on your competitor's actions
 - Change your costs
 - Employ strategies such as price matching instead of engaging in price wars
 - Lock in long-term contracts making price competition less attractive
 - Build excess capacity and signal willingness to flood the market if challenged
 - Create public announcements to signal future intentions
 - In repeated games look for trigger strategies. Knowing these can alter behavior.
 - For example, in the repeated prisoner's dilemma, both players cooperate initially, but one player defects for a period in response to the other player defecting (trigger)
 - Cooperate if possible – be careful about antitrust laws.

Key Takeaways

- Game theory models situations of interdependence: outcomes depend on others' actions.
- Nash equilibrium = no incentive to unilaterally deviate from strategy.
- In Prisoner's Dilemma, rational self-interest leads to worse group outcomes.
- Sequential games allow players to commit or observe and react.
- Credible commitments (e.g., price matching, ad campaigns) can shift game outcomes.
- In repeated games, cooperation is easier to sustain through:
 - Trigger strategies: punish defection
 - Tit-for-tat: start nice, retaliate, forgive
- Game of Chicken: commitment and reputation matter more than dominance.
- Smart firms learn to change the rules (entry deterrence, cost changes, signaling) to improve payoffs.

Knowledge Check

1. What is a Nash equilibrium?
2. What's the difference between sequential and simultaneous games?
3. Why is the Prisoner's Dilemma an example of rational decisions leading to bad outcomes?
4. What makes a threat "credible" in game theory?
5. How do repeated games help firms sustain cooperation?

Knowledge Check Answers

1. A set of strategies where no player can benefit from changing their action unilaterally.
2. In sequential games, players move in turns; in simultaneous games, players act at the same time.
3. Each player's dominant strategy leads to a worse outcome than cooperation would.
4. It must be observable, believable, and against the player's short-term interest.
5. They allow firms to punish defection and reward cooperation, sustaining better long-term outcomes.

Consultant Challenge's: Winning Strategic Games

Scenario:

- You're hired by a rideshare platform to help navigate pricing decisions in a highly competitive city. Their biggest rival keeps undercutting them, leading to pricing wars and driver churn.

Your Task:

1. How would you model the interaction between the two platforms? (Matrix? Tree?)
2. How can your client alter the structure of the game to make cooperation or stable pricing more likely?
3. Which strategies (commitments, signaling, repeated game discipline) would you recommend?